Recall: the Extended Delta Conjecture:

$$h_\ell[B]\, e_{m-\ell-1}[B-1]\, e_{N-\ell}(X) = \langle z^{N-m}\rangle \sum_{\substack{\lambda\vdash D_N \\ P\in L_{N,\ell}(\lambda)}} q^{\mathrm{dinv}(P)}\, t^{|\delta/\lambda|}\, x^{\mathrm{wt}_t(P)} \prod_{\substack{1<i\le N \\ c_i(\lambda)=c_{i-1}(\lambda)+1}} \left(1+zt^{-c_i(\lambda)}\right).$$

We proved:

Theorem 4.4.1: For $0\le \ell < m \le N$, we have

$$\left(\omega\left(h_\ell[B]\, e_{m-\ell-1}[B-1]\, e_{N-\ell}\right)\right)(x_1,\dots,x_m) = H_{q,t}^m\left(\frac{x_1x_2\cdots x_m}{\prod_{i=1}^m\left(1-qt\frac{x_i}{x_m}\right)}\, h_{N-m}(x_1,\dots,x_m)\, e_\ell(x_1^{-1},x_2^{-1},\dots,x_m^{-1})\right)_{\mathrm{pol}}$$

Theorem 5.1.1: For $0\le \ell < m \le N$, we have

$$\langle z^{N-m}\rangle \sum_{\substack{\lambda\vdash D_N \\ P\in L_{N,\ell}(\lambda)}} t^{|\delta/\lambda|} \prod_{\substack{1<i\le N \\ c_i(\lambda)=c_{i-1}(\lambda)+1}} \left(1+zt^{-c_i(\lambda)}\right) q^{\mathrm{dinv}(P)}\, x^{\mathrm{wt}_t(P)} = \sum_{\substack{J\subseteq[m-1] \\ |J|=\ell}} \sum_{\substack{\tau,(\vec{a},\vec{b})\in\mathbb{N}^m \\ |\tau|=N-m}} t^{|\vec{a}|}\, q^{d((\vec{a},\vec{b})+\tau)+h_J(\vec{a})}\, N_{\beta/\alpha}(X;q) \quad \text{where } \beta=(0,\vec{a})+(\underbrace{1,1,\dots,1}_{m})+\tau,\ \alpha=(\vec{a},0)+e_J$$

Hence the Extended Delta conjecture can be written as:

$$\omega\left(H_{q,t}^m\left(\frac{x_1x_2\cdots x_m}{\prod_{i=1}^m\left(1-qt\frac{x_i}{x_m}\right)}\, h_{N-m}(x_1,\dots,x_m)\, e_\ell(x_1^{-1},x_2^{-1},\dots,x_m^{-1})\right)_{\mathrm{pol}}\right) = \sum_{\substack{J\subseteq[m-1] \\ |J|=\ell}} \sum_{\substack{\tau,(\vec{a},\vec{b})\in\mathbb{N}^m \\ |\tau|=N-m}} t^{|\vec{a}|}\, q^{d((\vec{a},\vec{b})+\tau)+h_J(\vec{a})}\, N_{\beta/\alpha}(X;q) \quad \text{where } \beta=(0,\vec{a})+(\underbrace{1,1,\dots,1}_{m})+\tau,\ \alpha=(\vec{a},0)+e_J$$

i.e.

$$H_{q,t}^m\left(\frac{x_1x_2\cdots x_m}{\prod_{i=1}^m\left(1-qt\frac{x_i}{x_m}\right)}\, h_{N-m}(x_1,\dots,x_m)\, e_\ell(x_1^{-1},x_2^{-1},\dots,x_m^{-1})\right)_{\mathrm{pol}} = \sum_{\substack{J\subseteq[m-1] \\ |J|=\ell}} \sum_{\substack{\tau,(\vec{a},\vec{b})\in\mathbb{N}^m \\ |\tau|=N-m}} t^{|\vec{a}|}\, q^{d((\vec{a},\vec{b})+\tau)+h_J(\vec{a})}\, (\omega N_{\beta/\alpha})(X;q) \quad \text{where } \beta=(0,\vec{a})+(\underbrace{1,1,\dots,1}_{m})+\tau,\ \alpha=(\vec{a},0)+e_J$$

$$\|$$

$$H_q^m\left(\left(\prod_{1\le i<j\le m}\frac{\left(1-qt\frac{x_i}{x_j}\right)}{\left(1-t\frac{x_i}{x_j}\right)}\right)\frac{x_1x_2\cdots x_m}{\prod_{i=1}^m\left(1-qt\frac{x_i}{x_m}\right)}\, h_{N-m}(x_1,\dots,x_m)\, e_\ell(x_1^{-1},x_2^{-1},\dots,x_m^{-1})\right)_{\mathrm{pol}}$$

$$\|$$

$$H_q^m\left(\frac{\prod_{\substack{1\le i<j\le m \text{ s.t.} \\ i+1\le j}}\left(1-qt\frac{x_i}{x_j}\right)}{\prod_{1\le i<j\le m}\left(1-t\frac{x_i}{x_j}\right)}\, x_1x_2\cdots x_m\, h_{N-m}(x_1,\dots,x_m)\, \overline{e_\ell(x_2,\dots,x_m)}\right)_{\mathrm{pol}}$$

$$\therefore\ H_q^m\left(\frac{\prod_{\substack{1\le i<j\le m \text{ s.t.} \\ i+1\le j}}\left(1-qt\frac{x_i}{x_j}\right)}{\prod_{1\le i<j\le m}\left(1-t\frac{x_i}{x_j}\right)}\, x_1x_2\cdots x_m\, h_{N-m}(x_1,\dots,x_m)\, \overline{e_\ell(x_2,\dots,x_m)}\right)_{\mathrm{pol}} = \sum_{\substack{J\subseteq[m-1] \\ |J|=\ell}} \sum_{\substack{\tau,(\vec{a},\vec{b})\in\mathbb{N}^m \\ |\tau|=N-m}} t^{|\vec{a}|}\, q^{d((\vec{a},\vec{b})+\tau)+h_J(\vec{a})}\, (\omega N_{\beta/\alpha})(X;q)$$

$$\text{where } \beta=(0,\vec{a})+(\underbrace{1,1,\dots,1}_{m})+\tau,\ \alpha=(\vec{a},0)+e_J$$